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WALMART SALES PERFORMANCE ANALYSIS

WEEKLY SALES DRIVERS AND STORE PERFORMANCE
INSIGHTS

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PROJECT OVERVIEW

This project analyzes Walmart weekly sales data across multiple stores to understand sales performance and identify key drivers such as holidays and external economic factors. The goal is to generate insights that support decision-making in forecasting and store performance evaluation.

Key objectives

To analyze Walmart's weekly sales across multiple stores and time periods to **identify factors influencing sales performance** and highlight high and low performing stores.

Problems identified

- Raw data is not interpretable at a glance (no clear story in the dataset alone)
- Difficulty understanding what actually drives sales changes
- Hard to compare store performance without structured analysis
- No clear visibility on whether external factors (economic indicators, holidays, fuel price) matter significantly
- No summarized insights or dashboards for decision-making

Target Audience

- **Primary audience:** Walmart management and business stakeholders
- **Secondary audience:** Data/BI team or hiring managers
- **Tertiary audience:** Non-technical business users

User story

Walmart wants to analyze and understand store performance using historical retail data **to identify drivers of weekly sales and improve decision-making** around sales forecasting, store performance evaluation, and business planning

Use cases

1. Store performance analysis

Analyze weekly sales across all stores to identify top-performing and underperforming locations and compare performance over time.

Acceptance Criteria

The dashboard should:

- Show total weekly sales by store.
- Allow comparison of sales performance across all stores.
- Clearly highlight the top performing and lowest performing stores.
- Allow filtering by store and date.

2. Sales trend and seasonality analysis

Examine weekly sales patterns to identify trends, seasonal behavior, and periods of growth or decline.

Acceptance Criteria

The dashboard should:

- Display weekly sales trends over time.
- Allow users to view sales patterns across different time periods
- Clearly show periods of growth, decline, or unusual sales activity.

3. Holiday sales impact

Compare holiday weeks with non-holiday weeks to determine how holidays influence sales performance

Acceptance Criteria

The dashboard should:

- Distinguish between holiday and non-holiday weeks
- Compare average weekly sales during holiday weeks versus non-holiday weeks.
- Clearly show whether holidays lead to higher or lower sales.

4. External factor analysis

Evaluate how temperature, fuel prices, CPI, and unemployment relate to changes in weekly sales.

Acceptance Criteria

The dashboard should:

- Display the relationship between weekly sales and temperature, fuel price, CPI and unemployment.
- Indicate the direction of each relationship (positive or negative correlation) between the external factor and weekly sales
- Identify which external factor shows strongest association with weekly sales performance

5. Sales forecasting support

Use historical sales patterns and identified drivers to support future sales forecasting and planning.

Acceptance Criteria

The dashboard should:

- Present historical weekly sales patterns to **support forecasting analysis**
- Highlight seasonal peaks and troughs that recur across years to support planning cycle alignment
- Allow filtering by store and time period so analysts can generate store-specific historical baselines for forecasting inputs

Success criteria from user perspective

The solution is successful if it can:

- Easily identify top performing and underperforming stores based on weekly sales and act on store-level performance differences.
- Assess how sales trends, seasonality, and anomalies affect overall performance and use this to make informed **planning decisions**.
- Compare holiday and non-holiday sales performance to determine the impact of holidays on **revenue** and adjust operational strategies accordingly.
- Evaluate how external factors (temperature, fuel price, CPI, unemployment) influence sales and use these insights to support **pricing**, staffing, and planning decisions.
- Use historical **sales patterns** to support forecasting and **budgeting decisions** across stores and time periods.
- Interact with the dashboard by store and time period to quickly isolate issues or opportunities and support **faster decision-making**.

Information needed from user perspective

The information needed for Walmart to solve **business problems**.

Example:

- Weekly sales per store
- Performance comparison across stores
- Impact of holidays on sales
- Influence of economic factors (CPI, unemployment, fuel price, temperature)

Data needed from analyst perspective

The raw fields required to support these **business needs** include:

Example schema:

Field	Data Type
Store	String/Integer
Weekly Sales	Float
Date	Date
Temperature	Float
Fuel Price	Float
CPI	Float
Unemployment	Float
Holiday Flag	Boolean/Integer

Data quality checks

Measures to ensure the dataset is complete, accurate, and usable for analysis:

- **Row count check** – Confirm expected number of records (no missing batches of data)
- **Column check** – Ensure all required fields are present (Store, Date, Weekly_Sales etc.)
- **Data type validation** – Verify numeric, date, and categorical fields are correctly formatted
- **Duplicate check** – Ensure no duplicate rows across store + date combinations
- **Missing value check** – Identify and handle null or blank values in key fields)
- **Range validation** – Confirm values are realistic (e.g. No negative sales, valid CPI/unemployment ranges)
- **Date continuity check** – ensure weekly data has no missing time periods per store

Additional requirements

- Document the full solution including dataset description, transformations, and analysis approach
- Clearly explain data sources and how the dataset was prepared for analysis.
- Include step-by-step transformation process (cleaning, formatting, aggregation where needed)
- Provide final insights and conclusions derived from the analysis
- Build a reproducible workflow so the analysis can be rerun with updated data
- Maintain a clear project structure suitable for future updates and scaling
- Store project files, dashboards, and documentation in a version-controlled repository (e.g. GitHub)

